



added here again to regularize



smooth, convex optimization

Sinkhorn Algorithm for Barycenters

- Entropic smoothing can be **added here again to regularize**:

$$\alpha^\star = \arg \min_{\alpha} \sum_{i=1}^K \lambda_i \text{OT}_{\varepsilon}(\alpha, \beta_i)$$

- Again, yields a **smooth, convex optimization** problem
- Can be solved via Sinkhorn iterations. At step t :
 1. Iteratively update scaling vectors $(\mathbf{u}_s^{(t)}, \mathbf{v}_s^{(t)})$ for each s (parallelizable)
 2. Update $\mathbf{a}$ via $\mathbf{a}^{(t+1)} \leftarrow \prod_s (\mathbf{K}_s \mathbf{v}_s^{(t)})^{\lambda_s}$
- Advantages: uniqueness, smoother solutions, faster convergence.

Comparison: Classical vs Entropic

Classical barycenters

- Exact, preserves sharp features.
- Computationally heavy.

Entropic barycenters

- Approximate, smoother (blurred).
- Much faster and scalable.